A LIGHTWEIGHT ANPR SYSTEM

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Introduction

What is ANPR?

- ANPR (Automatic Number Plate Recognition) is a computer-vision system that automatically detects and reads vehicle license plates from images.
- Widely used in:
  - ✓ Smart traffic & security systems
  - ✓ Parking management
  - ✓ Toll collection
- Identifying number plates manually is inefficient for modern traffic and security systems, especially because traditional rule-based methods fail under varying lighting, angles, and image quality.
- An automated recognition method provides faster, more consistent, and more accurate results.

Objectives

A	To understand the fundamental steps of image preprocessing used in computer vision.
B	To detect the number plate region from a vehicle image using contour-based localization.
C	To extract and crop the detected plate area for further processing.
D	To perform optical character recognition (OCR) to read the alphanumeric text on the plate.
E	To verify the system's accuracy by testing on multiple images and analyzing results.

Methodology

Step 1: Image Acquisition

i. The user uploads an image from their device using *"files.upload()"*

Step 2: Preprocessing

i. Convert the image from BGR → **Grayscale**
ii. Apply **bilateral filter**
iii. Detect edges using **Canny operator** to reduce noise while preserving edges

Step 3: Contour Detection

i. Find all contours
ii. Select top Contours by area
iii. Approximate polygons and identify the one with 4 sides

How does Canny Edge Detection work?

- Reduces noise using Gaussian filter.
- Calculates image intensity gradients to detect edges.
- Suppresses non-edge pixels.
- Uses double thresholding and hysteresis to form continuous edges.
- Produces clean boundaries for object detection.

How does Contour Detection work?

- Finds continuous boundaries from Canny edge-detected images.
- Represents objects as connected point sequences.
- Contours are sorted by area.
- Polygon approximation identifies rectangular shapes.
- Four-sided contours are selected as number plates.

Methodology (Cont.)

Step 4: Masking & Cropping

i. Create a binary mask
ii. Extract only the region of interest (ROI)
iii. Crop tightly around the detected plate

Step 5: OCR Recognition

i. EasyOCR reads characters from cropped region
ii. Characters are joined into a single string

Step 6: Final Output

i. Recognized number is printed
ii. Plate area is highlighted on the original image

How does EasyOCR work?

- Uses deep learning to detect text regions.
- Crops detected regions automatically.
- Recognizes alphanumeric characters using neural networks.
- Outputs text with confidence scores.
- Works without manual training or templates.

Fallback Strategy

If plate contours are not detected in **Step-3**, OCR is applied directly to the full image in **Step-5** to avoid complete system failure

Result and Analysis

Original Image



Grayscale



Noise Reduction



Canny Operator



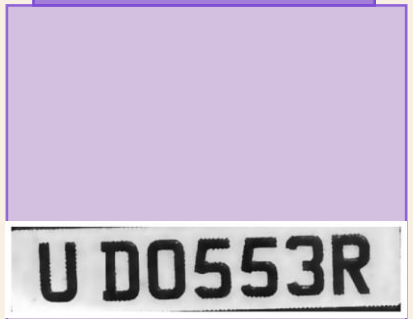
Contour Detection



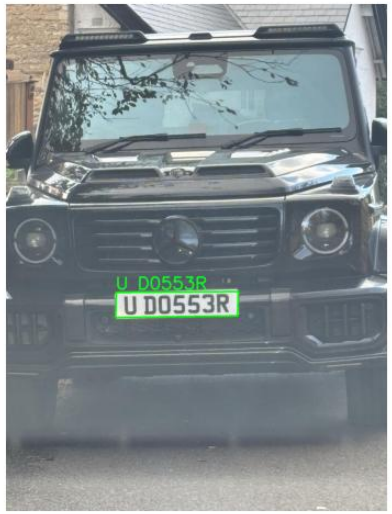
Plate Detection



Cropped Plate



Final Output



Result and Analysis (Cont.)

Key Observations:

- ✓ The system successfully detected the plate region.
- ✓ EasyOCR correctly recognized alphanumeric characters with high accuracy when the plate was distinct.
- ✓ The combination of bilateral filtering + Canny edge detection produced clean contours suitable for plate localization.

Limitations:

- ***Low-Light Performance:*** Accuracy decreases significantly for images captured in poor lighting or nighttime conditions.
- ***Image Quality Dependency:*** Blurred, low-resolution, or partially occluded plates can cause OCR misinterpretation.
- ***Lack of Skew Correction:*** The current system does not correct rotation or tilt, reducing accuracy for angled plates.
- ***Complex Background Sensitivity:*** Contour-based detection may fail when the background contains patterns similar to plate edges. Again, busy plates reduce accuracy as well.
- ***Language and Font Restriction:*** The system is not optimized for Bangla license plates or non-standard stylized fonts.

Result and Analysis (Cont.)

Low Light

Final Detected Number: HRZSEX6969



Low Quality

Final Detected Number:



Angular Distortion

Final Detected Number: 4010126



Complex Background

Final Detected Number: NGJ 072 AWS TUISLWS



Language

Final Detected Number: DAGESSREV



Conclusions

Summary

- ✓ A working ANPR system was implemented using only classical image processing + EasyOCR.
- ✓ Successfully detects number plates and extracts characters from vehicle images.
- ✓ Preprocessing and contour-based detection make the system lightweight and fast.

Comparisons

- Deep learning–based ANPR systems are more robust but require training and GPU resources.
- This lightweight approach is simpler, faster and ideal for small-scale applications.

Future Improvements

- Add deep-learning YOLO-based plate detection
- Improve OCR using CNN-based text recognition
- Add rotation correction and skew removal
- Add Bangla license plate recognition capability

THANK YOU

